

§ 9-4

1. 设 $z = \frac{y}{f(x^2 - y^2)}$, 其中 $f(u)$ 为可导函数, 求 $\frac{\partial z}{\partial x}$.

$$\text{解: } \frac{\partial z}{\partial x} = \frac{-y \cdot f'(x^2 - y^2) \cdot 2x}{f^2(x^2 - y^2)} = \frac{-2xy f'(x^2 - y^2)}{f^2(x^2 - y^2)}$$

2. 设 $z = (3x - 2y)^{3x - 2y}$, 求 dz .

$$\begin{aligned} \text{解: } dz &= d e^{(3x-2y)\ln(3x-2y)} \\ &= e^{(3x-2y)\ln(3x-2y)} \cdot d[(3x-2y)\ln(3x-2y)] \\ &= (3x-2y)^{3x-2y} \cdot [(3x-2y) \cdot d\ln(3x-2y) + \ln(3x-2y) d(3x-2y)] \\ &= (3x-2y)^{3x-2y} \left[(3x-2y) \cdot \frac{1}{3x-2y} d(3x-2y) + \ln(3x-2y) d(3x-2y) \right] \\ &= (3x-2y)^{3x-2y} [1 + \ln(3x-2y)] \cdot (3dx - 2dy) \end{aligned}$$

上面是直接求微分二方法来做的。如果你不习惯或不是那么熟悉微分二方法，就先求出 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$, 再写出 $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$. 例中有一题 $z = (1+xy)^y$, 本题是它的加强版。

3. 设 $z = f(e^{xy}, x^2 - y^2)$, 求 $\frac{\partial^2 z}{\partial x \partial y}$.

$$\begin{aligned} \text{解: } \frac{\partial z}{\partial x} &= f_1' \cdot e^{xy} \cdot y + f_2' \cdot 2x \\ \therefore \frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial y} (f_1' \cdot y e^{xy} + 2x f_2') \quad (\text{其中 } f_1' = f_1'(e^{xy}, x^2 - y^2); f_2' = f_2'(e^{xy}, x^2 - y^2)) \\ &= y e^{xy} [f_{11}'' \cdot e^{xy} \cdot x + f_{12}''(-2y)] + f_1' \cdot e^{xy} + f_1' \cdot y \cdot e^{xy} \cdot x \\ &\quad + 2x [f_{21}'' e^{xy} \cdot x + f_{22}''(-2y)] \\ &= (1 + xy e^{xy}) f_{11}' + xy e^{2xy} f_{11}'' + 2(x^2 - y^2) e^{xy} f_{12}'' - 4xy f_{22}'' \\ &\quad (\text{若 } f \text{ 具有二阶连续偏导}) \end{aligned}$$

§ 9-5

1. 设 $x = x(y, z)$, $y = y(z, x)$, $z = z(x, y)$ 都是由方程 $F(x, y, z) = 0$ 所确定的隐函数,

则下列等式中, 不正确的是 (C).

(A) $\frac{\partial x}{\partial y} \cdot \frac{\partial y}{\partial x} = 1$; $-\frac{F_y}{F_x} \cdot -\frac{F_x}{F_y} = 1$ ✓ (B) $\frac{\partial x}{\partial z} \cdot \frac{\partial z}{\partial x} = 1$; $-\frac{F_z}{F_x} \cdot (-\frac{F_x}{F_z}) = 1$. ✓

(C) $\frac{\partial x}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x} = 1$; (-1) (D) $\frac{\partial x}{\partial y} \cdot \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x} = -1$. ✓

$-\frac{F_y}{F_x} \cdot -\frac{F_z}{F_y} \cdot -\frac{F_x}{F_z} = -1$. ✓

2. 设 $z = z(x, y)$ 是由方程 $xy - yz + zx = e^z$ 所确定的隐函数, 求该函数在 (1, 1) 点处的偏导数.

解: $F(x, y, z) = xy - yz + zx - e^z$. | 12 |

$F_x = y + z$; $F_y = x - z$; $F_z = -y + x - e^z$.

$\therefore \frac{\partial z}{\partial x} \Big|_{\substack{x=1 \\ y=1}} = -\frac{F_x}{F_z} \Big|_{\substack{x=1 \\ y=1}} = -\frac{y+z}{-y+x-e^z} \Big|_{\substack{x=1 \\ y=1}} = 1$ (其中 $z=0$, 将 $x=1, y=1$ 代入原方程可得)

$\frac{\partial z}{\partial y} \Big|_{\substack{x=1 \\ y=1}} = -\frac{F_y}{F_z} \Big|_{\substack{x=1 \\ y=1}} = \frac{z-x}{-y+x-e^z} \Big|_{\substack{x=1 \\ y=1}} = 1$.

3. 求由方程组 $\begin{cases} z = x^2 + y^2, \\ x^2 + 2y^2 + 3z^2 = 20 \end{cases}$ 所确定的函数的导数 $\frac{dy}{dx}$ 和 $\frac{dz}{dx}$.

解: $F(x, y, z) = x^2 + y^2 - z$
 $G(x, y, z) = x^2 + 2y^2 + 3z^2 - 20$

$\therefore F_x = 2x, F_y = 2y, F_z = -1$.
 $G_x = 2x, G_y = 4y, G_z = 6z$.

$\therefore J = \frac{\partial(F, G)}{\partial(y, z)} = \begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix} = \begin{vmatrix} 2y & -1 \\ 4y & 6z \end{vmatrix} = 4y(3z+1)$.

$\frac{\partial(F, G)}{\partial(x, z)} = \begin{vmatrix} F_x & F_z \\ G_x & G_z \end{vmatrix} = \begin{vmatrix} 2x & -1 \\ 2x & 6z \end{vmatrix} = 2x(6z+1)$.

$\frac{\partial(F, G)}{\partial(y, x)} = \begin{vmatrix} F_y & F_x \\ G_y & G_x \end{vmatrix} = \begin{vmatrix} 2y & 2x \\ 4y & 2x \end{vmatrix} = -4xy$.

$\therefore \frac{dy}{dx} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, z)} = -\frac{x(6z+1)}{2y(3z+1)}$; $\frac{dz}{dx} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, x)} = \frac{x}{3z+1}$.